

Unit 12, Lesson 11: Solving One-Step Multiplication and Division Equations Using Diagrams

Benchmark

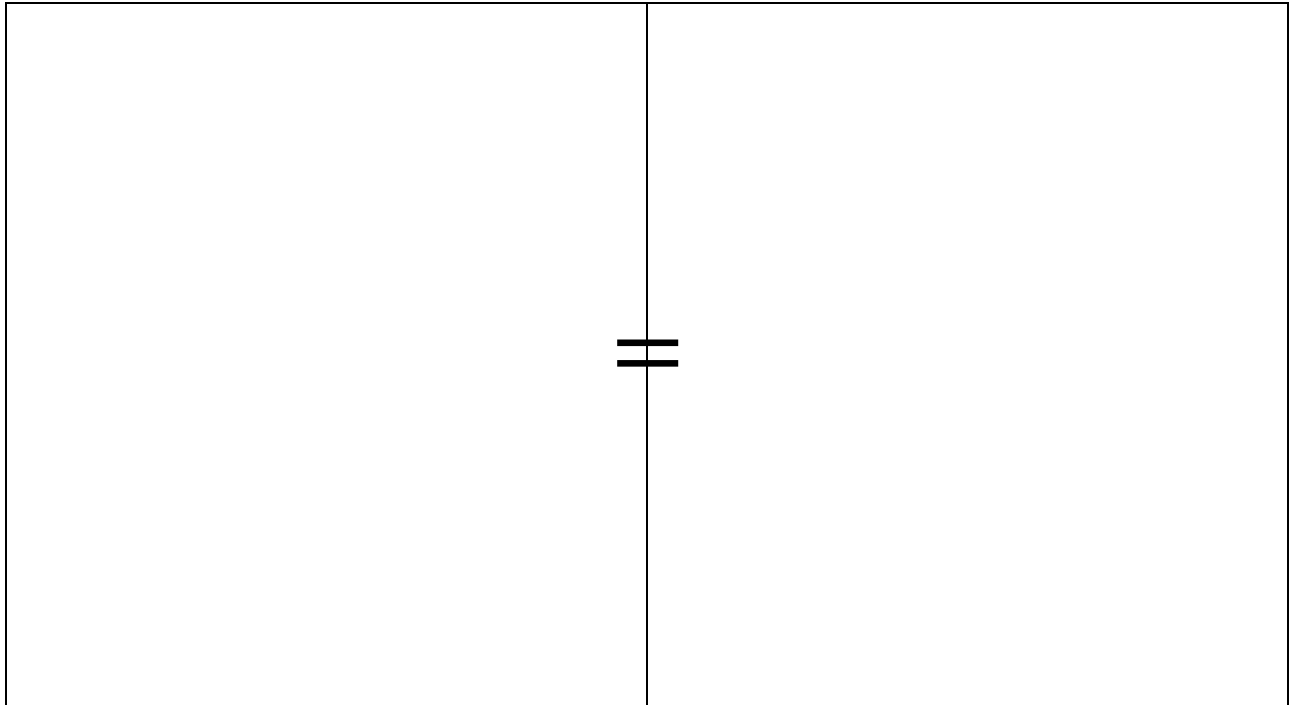
MA.6.AR.2.3 Write and solve one-step equations in one variable within a mathematical or real-world context using multiplication and division, where all terms and solutions are integers.

Learning Target

- ☐ I can solve one-step multiplication and division equations using diagrams.

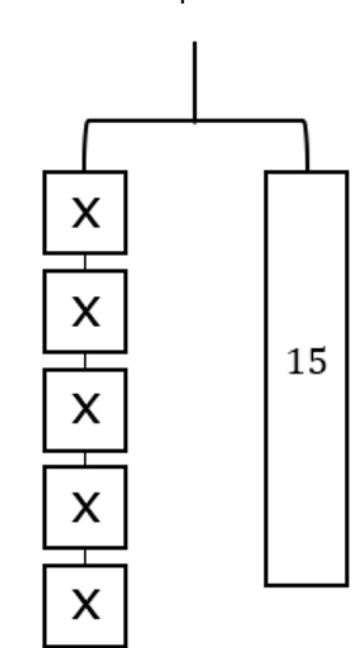
12.11.1 Warm-Up: Manipulative Math

1. On the equation mat, sketch algebra tiles to represent the equation $7x = -14$, and show how to find the solution.



12.11.2 Exploration: Hanging Around Activity

1. The hanger diagram represents the equation $5x = 15$.



- a. Complete the statements.

The left side of the hanger is broken into
equal parts. If 15 is broken into the same number of
equal parts, each part will have the same weight as
one x block. Splitting each side of the hanger into

parts is the same as ☐ dividing
☐ multiplying each

side of the equation by

- ☐ 3
☐ 5
☐ 15

- ☐ 3
☐ 5
☐ 15

- ☐ 3
☐ 5
☐ 15

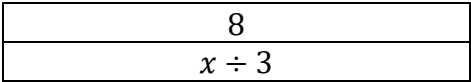
- b. Sketch splitting the hanger diagram into the equal parts as described in Part A.
- c. What is the value of x ?
- d. Compare your diagram with your partner. Discuss the similarities and differences.
- e. Provide positive feedback to your partner in their book on their sketch and solution.

Partner Feedback	
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12.11.3 Guided Instruction: Solving One-Step Multiplication and Division Equations Using Diagrams

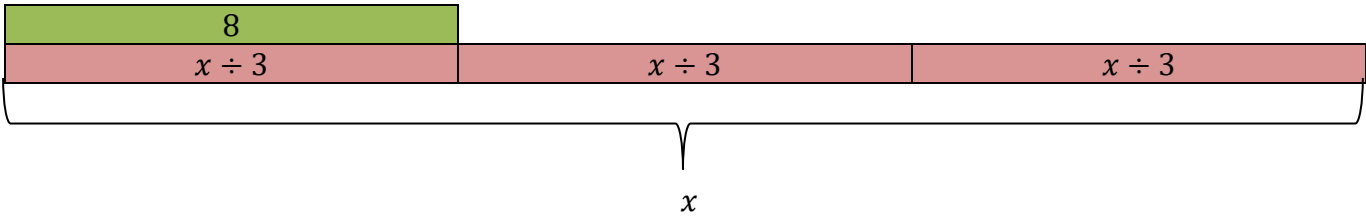
Tape diagrams, also known as bar models, can be used to model and solve multiplication and division equations.

1. The equation $\frac{x}{3} = 8$ is represented by the tape diagram shown.



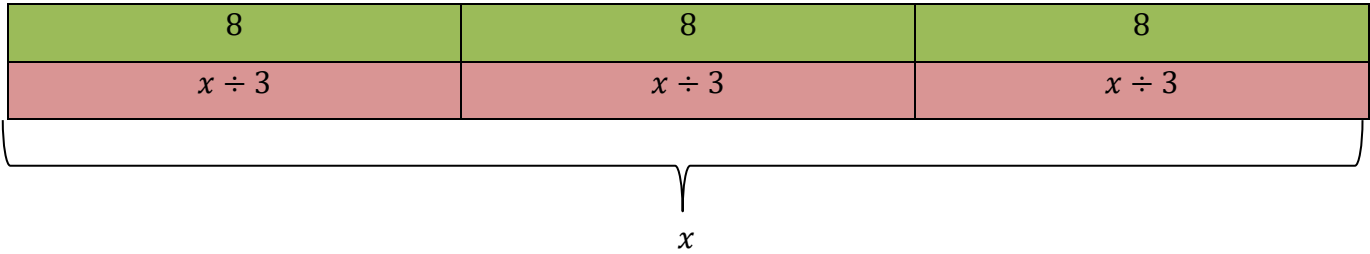
- a. The solution to the equation $\frac{x}{3} = 8$ can be modeled in the following steps. Work with your partner to explain what is occurring in each step.

Step 1



Explanation of Step 1

Step 2



Explanation of Step 2

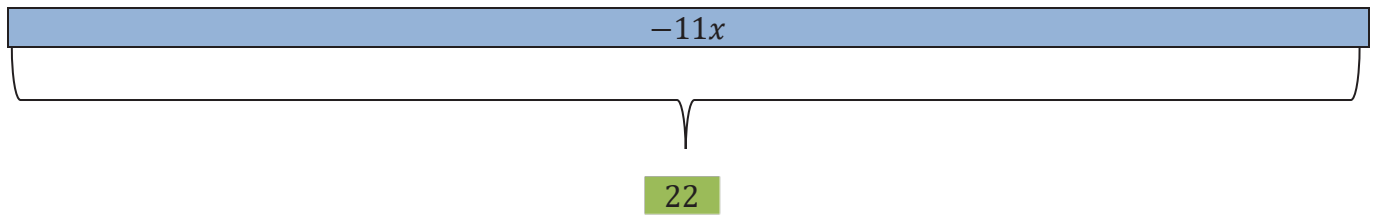
Step 3

24
x

Explanation of Step 3

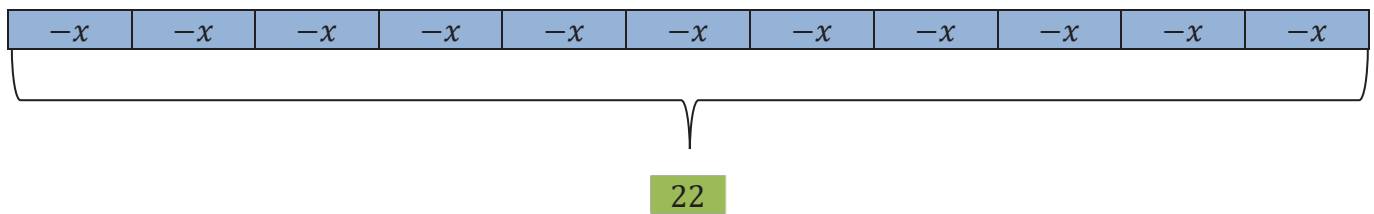
- b. Discuss with your partner how the operation used in the equation relates to the operation used in the solution. Summarize your discussion.

2. The tape diagram models the equation $22 = -11x$.



The solution to the equation $22 = -11x$ can be modeled in the following steps. Work with a different partner to explain what is occurring in each step.

Step 1



- a. How were the integers in the equation used to determine the negative value of x ?

Explanation of Step 1	

Step 2

$$\underbrace{-x}_2$$

Explanation of Step 2**Step 3**

$$\underbrace{x}_{-2}$$

Explanation of Step 3

- b. Discuss with your partner how the operation used in the equation relates to the operation used in the solution. Summarize your discussion.

12.11.4 Collaboration: Matching Equations and Solutions

1. Match each equation to the correct solution. Record the equation with its solution in the table. Some solutions may be used more than once or not at all. Include an explanation for the match.

A. $x = 2$	E. $x = 3$
B. $x = -20$	F. $x = 10$
C. $x = -10$	G. $x = -8$
D. $x = 9$	H. $x = -3$

1. $-3x = -6$

2. $-\frac{x}{2} = 4$

3. $\frac{x}{3} = 3$

4. $-\frac{x}{2} = 5$

5. $3x = 9$

6. $\frac{x}{10} = -2$

7. $4x = 8$

Equation	Solution	Explanation for Match

12.11.5 Your Turn!

Sketch a tape diagram to represent each equation. Then, solve the equation.

1. $-5p = -40$

2. $-2 = \frac{m}{7}$

3. $-6 = \frac{w}{9}$

12.11.6 Wrap-Up: Ticket Out the Door

1. Throughout this unit, you have been shown how to solve equations using tape diagrams, hanger diagrams, number lines, manipulatives, and inverse operations. Use one of these methods to model and solve the equation $6h = -12$.

